AUSTRALIAN RESEARCH COUNCIL
Discovery Early Career Researcher Award
Application for Funding Commencing in 2027

DE

Project ID: DE270101085

First Investigator: Dr Sorel De Leon Vergara

Admin Org: RMIT University

Total number of sheets contained in this Application: 29

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Part A - Administrative Summary (DE270101085)

A1. Application Title

(Provide a short title. (Up to 75 characters, approximately 10 words).)

Peripheral nerves: A new paradigm in brain-machine interfaces

A2. Person Participant Summary

(Add the participant to this application.)

Number	Name	Participant Type	Current Organisation(s)
1	Dr Sorel De Leon Vergara	Discovery Early Career Researcher Award	RMIT University

A3. Organisation Participant Summary

(Add the Administering Organisation participating in this application. Refer to the Instructions to Applicants for further information.)

Number	Name	Participant Type
1	RMIT University	Administering Organisation

A4. Application Summary

(Provide an Application Summary, focusing on the aims, significance, expected outcomes and benefits of this project. Write the Application Summary simply, clearly and in plain English. If the application is successful, the Application Summary, along with the National Interest Test, will be used to give the general community an understanding of the research. Avoid the use of acronyms, quotation marks and upper-case characters. Refer to the Instructions to Applicants for further information. (Up to 750 characters, approximately 100 words).)

This project aims to improve understanding of how electrical signals used in brain–machine interfaces are recorded from the body by studying the connection between nerves and muscle. By combining computer modelling with laboratory experiments, the research will examine how electrode design and placement affect signal quality and stability during recording and stimulation from nerve-muscle interfaces. The project will develop realistic 3D nerve-muscle models to identify physical limits in existing technologies. Expected outcomes include new electrode design guidelines and open digital tools for researchers. This work will support the future development of safer, more reliable technologies for controlling prosthetic and assistive devices.

A5. National Interest Test Statement

(See the Instructions to Applicants for addressing the National Interest Test and 'Articulating National Interest in Grant Applications' available on the ARC website (up to 1,500 characters, approximately 200 words).)

Today, medical devices can read electrical signals from the body, yet these technologies often struggle to capture these signals reliably over long periods. This project addresses this gap by improving understanding of how signals from nerves and muscles are generated and recorded by the small sensors used in brain–machine interfaces. By combining laboratory experiment with computer modelling, the research will examine how sensor design, placement, and materials affect signal quality and long-term stability.
The project will develop realistic nerve–muscle models and open-source digital tools to help researchers and industry design more reliable sensing technologies. These advances could support the development of safer and more effective prosthetic and assistive devices, improving quality of life for people with disability and strengthening Australia’s growing medical device sector. The work aligns with national priorities in advanced manufacturing, artificial intelligence, and biotechnology, contributing to economic growth and reinforcing Australia’s position in the rapidly expanding neurotechnology field.
To maximise impact beyond academia, results will be shared through open-source resources, collaboration with industry partners, and engagement with clinical and engineering communities to accelerate uptake. This will support translation into new products, and enable broader adoption of safer human-machine interface technologies.

Part B - Participant Details including ROPE (Dr Sorel De Leon Vergara)

B1. Personal Details

(To update any Personal Details, click on the 'Manage Personal Details' link below. Note this will open a new browser tab. When returning to the form ensure to 'Refresh' the page to capture the changes made to the participant's profile.
Note: The date of birth, country of birth, citizenship, material personal interests and Indigenous status section will not appear in the PDF version of the form and will not be visible to assessors.
Data may be shared with other Commonwealth Entities.

All information contained in Part B is visible to the Administering Organisation on this application.)

Participation Type

Discovery Early Career Researcher Award

Title

Dr

First Name

Sorel

Middle Name

Elena

Family Name

De Leon Vergara

B3. Qualifications

(To update any qualifications, click on the 'Manage Qualifications' link below. Note this will open a new browser tab. When returning to the form ensure to 'Refresh' the page to capture the changes made to the participant's profile.)

Conferral Date	AQF Level	Degree/Award Title	Discipline/Field	Awarding Organisation	Country of Award
17/01/2022	Doctoral Degree	Doctor of Philosophy	Biomedical Engineering	Swinburne University of Technology	Australia
05/07/2017	Masters Degree	Master of Science	Bioelectronics and Nanotechnology	Maastricht University and Hasselt University	Belgium
21/02/2014	Bachelor Honours Degree,Graduate Certificate, Graduate Diploma	Professional Engineer	Electronic Engineer	Universidad Pontificia Bolivariana	Colombia

B4. Research Load (non-ARC Grants and Research)

(Provide details of research funding from non-ARC sources (in Australia and overseas). For research funding from non-ARC sources, list all projects/applications/awards/fellowships awarded or requests submitted involving that participant for funding for the years 2026 to 2030 inclusive.)

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Description (All named investigators on any application or grant/fellowship in which the participant is involved, project title, source of support, scheme and round)	**Same Research Area** [Yes (Y)/No (N)]	**Support Status** [Requested (R)/Current (C)/Past(P)]	**Application/Project ID** (for NHMRC applications only)	**2026** $'000	**2027** $'000	**2028** $'000	**2029** $'000	**2030** $'000
Dr Sorel De Leon Vergara, Rana Jaylani, Sian McConchie; Simulating the optimal brain machine interface in COMSOL; RACE Merit Allocation Scheme (RMAS), 2026	Y	C	N/A	3	0	0	0	0
Prof Robert Kapsa, Assoc Prof David Garrett, Dr Anita Quigley, Prof Denny Oetomo, Prof Elena Pirogova, Prof Simon Moulton, Dr Shaun Cloherty, **Dr Sorel De Leon Vergara**, Nethmini Walliwala Gamage; Electro-Neural Adaptive Bionic Living Electrode-interface (ENABLE-i); NHRMC Ideas Grants, 2024	Y	C	2038179	566	566	0	0	0

B5. Research Opportunity and Performance Evidence (ROPE) - Current and previous appointment(s) / position(s) - during the past 10 years

(To update any details in this table, click on the 'Manage Employment Details' link in this question. Note this will open in a new browser tab. 'Refresh' the application page when returning to the form to capture changes made to the participant's profile.)

Description	Department	Contract Type	Employment Type	Start Date	End Date	Organisation
Research Fellow	Biomedical Engineering	Contract	Full Time	28/05/2025	28/05/2028	RMIT University
Postdoctoral researcher	School of Engineering	Contract	Full Time	24/05/2021	27/05/2025	RMIT University
Research Assistant	Engineering	Contract	Full Time	16/07/2017	20/10/2017	Hasselt University, Belgium

B6. Research Opportunity and Performance Evidence (ROPE) - Career Interruptions

(You must read the ROPE Statement https://www.arc.gov.au/about-arc/program-policies/research-opportunity-and-performance-evidence-rope-statement before filling out this section.
Note - The following fields will not be visible to assessors: From when; To when; FTE of career interruption and Interruption category. RMS will automatically calculate the total career interruption in the field 'Total Period of Career Interruptions' which will be visible to assessors.

This question does not relate to the participant's eligibility for the scheme.)

Has the participant experienced a significant interruption that has impacted on research opportunity?

No

Total Period of Career Interruptions

B7. Research Opportunity and Performance Evidence (ROPE) – Career Narrative

(Provide an overview of the participant's career opportunities for research, including any periods of research only employment, access to periods of research leave, and proportions of time allocated for research, teaching and administrative responsibilities (Up to 565 characters, approximately 75 words).)

My PhD, conducted at CSIRO, was awarded in 2022 by Swinburne University of Technology and resulted in two publications. Following my PhD, I undertook full-time postdoctoral training at RMIT University, working in close collaboration with CarbonCybernetics. This postdoctoral work has led to two grants, six papers (including one as corresponding author), and a patent filing, as well as the co-supervision of four PhD students and supervision of two capstone projects. I have also served as a reviewer for Ideas Grants and top-tier journals (impact factor ~20).

B8. Research Opportunity and Performance Evidence (ROPE) - Evidence of the participant's research impact and contributions to the field, including those most relevant to this application

(Provide details of the participant's research impact and contributions. This should not include information provided elsewhere in the application (upload a PDF of 1 A4 page).)

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EVIDENCE OF RESEARCH IMPACT AND CONTRIBUTIONS TO THE FIELD

My research contributions have advanced the field of bioelectronic interfaces by establishing new methodologies and design principles for interfacing electrodes with complex biological tissues, particularly **three-dimensional (3D) cultures and neural systems**. A central and sustained contribution of my work has been the development of interface-centric approaches that link electrode geometry, materials, and impedance behaviour to biologically meaningful electrophysiological readouts.

Pioneering contributions to the application of impedance spectroscopy for 3-dimensional organotypic cultures

During my PhD, I led pioneering work on integrating electrodes with three-dimensional (3D) tissue cultures, at a time when most bioelectronic measurements were limited to two-dimensional monolayers. I developed and validated impedance-based physiological metrics that enabled non-destructive monitoring of 3D tissue maturation, viability, and responses to pharmacological perturbations. This work culminated in one review and one research article on impedance monitoring of 3D cultures (*BB 2020, IF 3.6; B MDPI 2021, IF 3.9*). Together, these papers have **accrued more than 90 citations to date and continue to show a steady citation rate**, with a **Field-Weighted Citation Impact of 1.21**. Importantly, this work has been cited in high-impact journals including *Nature Biomedical Engineering (IF 29)* and *ACS Chemical Reviews (IF 55)*, indicating its foundational nature and current relevance. I drove the conceptual framing, synthesis, and analysis of this work, which has since become a widely referenced entry point for researchers adopting impedance-based methods in tissue engineering and organotypic models.

This contribution helped establish **impedance spectroscopy as a credible and scalable tool for monitoring 3D biological systems**, supporting its uptake across drug testing, tissue engineering, and bioelectronics. The paper's sustained citation performance reflects its ongoing relevance and foundational role in the field.

Advancing electrode-cell interface theory and practice

Building on this foundation, my post-PhD research has delivered substantive advances in **understanding and optimising the electrode–cell interface for neural and neuromuscular** applications. I developed carbon fibre electrode architectures and demonstrated their capacity for high-quality single-unit neural recordings, while also identifying critical limitations in electrode yield and long-term stability. Through a combination of computational modelling and experimental validation, I clarified how electrode proximity, geometry, and impedance properties govern signal amplitude and fidelity, generating design principles that extend beyond a single device platform. This work is currently in preparation for publication, with me as last and corresponding author, and the associated modelling framework has been released as **open-source code** (https://github.com/sodeve19/NEURON_in_COMSOL), enabling other researchers to directly apply and adapt the approach in the development of brain-machine interface prototypes.

My impact in the brain-machine interface field is further evidenced by multiple **high-impact publications** on carbon-based device fabrication and neural interfacing (*DRM 2023, IF 5.1; ACS AMI 2024, IF 8.5; BTM 2024, IF 5.7; AHM 2025, IF 10.8*), **a filed patent** covering a device to aid insertion of carbon fibre arrays into the brain, and additional work on **platinum-based electrodes for retinal stimulation** (*JNE 2025, IF 4.5*). These studies are increasingly cited together, reflecting their collective contribution to a growing subfield focused on scalable, minimally invasive bioelectronic interfaces.

The influence of this work has led to sustained international collaborations with the University of Sherbrooke (Canada) and the University of Michigan (USA). I was invited to serve as a guest co-editor for Frontiers in Neurology alongside Dr. Julianna Richie (University of Michigan), Prof. James Weiland (University of Michigan), and Dr. Wei Tong (University of Melbourne). In parallel, my collaboration with CarbonCybernetics has contributed to the company securing **$500K in Genesis funding,** to be recognised as a finalist in the **Australian Museum Eureka Prizes for Cross-disciplinary work**, and obtaining **Human Research Ethics** approval from the Royal Melbourne Hospital in 2025 - marking a significant translational milestone for carbon fibre-based brain-machine interface technologies.

Research impact, citation trajectory, and leadership

My publication record demonstrates a clear upward trajectory in impact, with consistent citation growth across both early foundational work and more recent device-focused studies. Importantly, this impact is driven by papers where I played a leading intellectual role, including first-, and corresponding-authored publications.

Collectively, my work has contributed to a shift in emphasis within the field from predominantly algorithmic compensation strategies toward a more **fundamental, interface-centric understanding of electrode performance**. This has influenced how electrodes are designed, characterised, and evaluated in both neural and three-dimensional tissue contexts. These contributions directly underpin my current research direction focused on neuromuscular junction interfaces and provide a strong foundation for sustained international research impact.

B9. Research Opportunity and Performance Evidence (ROPE) - Mentoring and Leadership

(Provide evidence of the participant's emerging leadership and mentoring capability, including details of any experience and achievements in research training and supervision (up to 750 characters, or approximately 100 words).)

I currently co-supervise four PhD students (2 last-year & 2 first-year). Previously, I have supervised 2 MSc students, 2 Capstone projects, and 1 BSc student. Two of my Capstone students were awarded Top Project Overall at the 2025 RMIT EnGenius event, with one continuing under my supervision as a PhD candidate at RMIT University. A MSc student I supervised from the University of Edinburgh has since commenced a PhD at the University of Melbourne. I established a fortnightly publication-planning meeting that has led to multiple manuscript drafts and papers under submission. I also serve as the departmental representative on the School of Engineering ECR Committee at RMIT, supporting engagement and mentoring of early career researchers.

B10. Research Opportunity and Performance Evidence (ROPE) - Research Outputs Context

(Provide clear information that explains the relative importance of the participant's research outputs in disciplinary context. Explain how the participant's track record in relation to relevant discipline conventions demonstrates their capability for outstanding research in the field of this application. (Up to 1,500 characters, approximately 200 words).)

I am an electronics engineer focused on bridging the gap between electronics and biology. My research centres on electrode–tissue interfaces for brain–machine systems, with emphasis on materials, device design, and electrophysiological performance. In this field, animal-based validation often extends development timelines, and research impact is demonstrated through peer-reviewed publications, open-source frameworks and patents. Authorship conventions typically assign the majority of experimental work and manuscript preparation to the 1st author, while the corresponding author leads the conceptual framework; equal first/last authorships are common and of equal merit. I have authored 7 peer-reviewed publications in high-impact journals, including ACS Applied Materials & Interfaces, Advanced Healthcare Materials, Bioengineering & Translational Medicine, and Diamond and Related Materials, with 1 preprint under review in Advanced Functional Materials and 2 additional manuscripts under review in Small and Analytica Chimica Acta. I am 1st author on 3 published papers and on the 3 manuscripts currently under review, and corresponding author on 2. My work emphasises rigorous interface characterisation, electrode fabrication, and translational relevance in brain-machine interface research. Several studies are increasingly cited together, indicating a coherent contribution to scalable, minimally invasive neural interfaces, supported by a filed patent and open-source modelling tools.

B11. Research Opportunity and Performance Evidence (ROPE) – 10 Career-Best Research Outputs

(Provide up to 10 career-best research outputs and provide clear information regarding their research impact and excellence. Mark the research outputs that are most relevant to this project categorised under the following headings: Authored books; Edited books; Book chapters; Refereed Journal articles; Fully refereed conference proceedings; Additional research outputs (including non-traditional research outputs and preprints or comparable resources). CVs and theses should not be included in this list. Each text box allows up to 150 characters, approximately 20 words annotation per output.)

Research Outputs Listing

Generated research output document follows on the next page

Ten Career-Best Research Outputs

[1] * Higham, Simon, Jung, Young Jun, De Leon, Sorel E., Tong, Wei & Au, Shun Long Cyril et al. 2026, 'Carbon Cybernetics Array: a miniaturized carbon-based microelectrode array for intracortical recording', doi:10.64898/2026.01.06.696090 (Additional Research Output)

Research Outputs Annotation This pre-print develops-validates a carbon-fibre electrode array for BMIs; I was co-first author, leading electrode functionalisation and validation.

[2] * De León, Sorel E., Higham, Simon, Jung, Young Jun, Tong, Wei & Garrett, David J. 2024, 'Recent developments in microwire-structured intracortical electrode arrays for brain–machine interfaces', *Bioengineering & Translational Medicine*, doi:10.1002/btm2.10742 (Refereed Journal Article)

Research Outputs Annotation This paper reviews advances in microwire-based electrodes; I am first and corresponding author, leading analysis and manuscript preparation.

[3] * De León, SE, Pupovac, A & McArthur, SL 2020, 'Three-Dimensional (3D) cell culture monitoring: Opportunities and challenges for impedance spectroscopy.', *Biotechnology and Bioengineering*, vol. 117, no. 4, pp. 1230–1240, doi:10.1002/bit.27270 (Refereed Journal Article)

Research Outputs Annotation This paper reviews electrode design for impedance monitoring of 3D cell cultures; I was first author and primary manuscript lead.

[4] * De Leon, SE, Cleuren, L, Oo, ZY, Stoddart, PR & McArthur, SL 2021, 'Extending In-Plane Impedance Measurements from 2D to 3D Cultures: Design Considerations', *Bioengineering*, vol. 8, no. 1, pp. 11, doi:10.3390/bioengineering8010011 (Refereed Journal Article)

Research Outputs Annotation This paper implements impedance monitoring in 3D cell cultures - identifying key limitations; I was first author and primary manuscript lead.

[5] * Higham, SJ, Rojas Cabrera, JM, Kwak, Y, Hong, L & Chambers, A et al. 2025, 'Boron-Doped Nano-Crystalline Coated Carbon Fibers for Phasic Dopamine Sensing.', *Advanced Healthcare Materials*, pp. e03945, doi:10.1002/adhm.202503945 (Refereed Journal Article)

Research Outputs Annotation This paper implements a boron-based coating on carbon-fibre electrodes for BMIs; I contributed to experimental design and lab work.

[6] * Falahatdoost, S, Prawer, YDJ, Peng, D, Chambers, A & Zhan, H et al. 2024, 'Control of Neuronal Survival and Development Using Conductive Diamond', *ACS Applied Materials & Interfaces*, vol. 16, no. 4, pp. 4361–4374, doi:10.1021/acsami.3c14680 (Refereed Journal Article)

Research Outputs Annotation This paper demonstrates that diamond coatings can enhance neural growth. I contributed to experimental design and lab work.

[7] * Spencer, Martin J, Hosie, Suzanne, Tong, Wei, Shivdasani, Mohit N & Garrett, David J et al. 2025, 'Towards a closed loop retinal prosthesis: measuring electrically evoked retinal responses using large electrodes', *Journal of Neural Engineering*, vol. 22, no. 4, pp. 046054, doi:10.1088/1741-2552/adfc9b (Refereed Journal Article)

Research Outputs Annotation This paper reports large platinum electrodes for retinal stimulation and recording; I contributed to electrode design and fabrication.

[8] * Khatereh Edalati, Simon Higham, Sorel E. De Leon, Shun Long Cyril Au & Athavan Nadarajah et al. 2023, 'Effective removal of metal-carbide contamination from diamond circuit boards by reactive ion etching', *Diamond and Related Materials*, vol. 139, pp. 110329, doi:10.1016/j.diamond.2023.110329 (Refereed Journal Article)

Research Outputs Annotation This paper reports reactive ion etching for contaminant removal in diamond circuits; I contributed to experimental design and manuscript writing.

B12. Currently held ARC Projects

(This information is automatically populated. If you have any concerns with the information recorded here, please contact your Administering Organisation's Research Office.)

B13. Relevant Qualification

(To update any qualifications, click on the 'Manage Qualifications' link in the Qualifications question above. This will open in a new browser tab. When returning to the form ensure to 'Refresh' the page to capture changes made to the participant's profile.)

Degree/Award Title	Awarding Organisation	Conferral Date
Doctor of Philosophy	Swinburne University of Technology	17/01/2022

B14. Has the participant been granted an extension by the Administering Organisation to the eligibility period due to a significant career interruption as outlined in the Grant Guidelines?

(If the participant's qualification relevant to this application (listed in Question B13) was awarded prior to 1 March 2021 and they have had a significant career interruption, the participant will need to seek an extension to the eligibility period through their Deputy Vice-Chancellor (Research).

If 'Yes' Questions B15 and B16 will be enabled and you will be required to specify the type(s) of career interruption claimed and the total period of the extension claimed.)

No

B15. Select the category of career interruption claimed (more than one may be selected)

(Choose all types of career interruptions which have been claimed in the application for extension to the participant's qualification as certified by the Deputy Vice-Chancellor (Research) or equivalent.
Select a type of interruption and click 'Add'.)

B16. What is the total period of extension that the participant has claimed?

(Select the period of time which most closely equals the total period of extension claimed.)

Part C - Project Description (DE270101085)

C1. Project Description

(Upload a Project Description as detailed in the Instructions to Applicants and in the required format. (Up to 7 A4 pages including references).)

Uploaded PDF file follows on next page.

PROJECT TITLE: Peripheral nerves: A new paradigm in brain-machine interfaces

PROJECT QUALITY AND INNOVATION

Problem

Rapid advances in artificial intelligence have accelerated demand for brain–machine interfaces (BMIs), yet current commercial systems remain functionally limited. While several technologies have entered human trials[1–3], most are restricted to low-bandwidth tasks such as cursor control and fail to translate user intention into natural, coordinated movement. Critically, these invasive technologies do not provide meaningful sensory or proprioceptive feedback. Without feedback, **closed-loop control essential for naturalistic prosthetic function**, cannot be achieved[4–6].

The primary **barrier is biological rather than computational**. Invasive brain implantation causes tissue damage and triggers inflammatory and fibrotic responses that progressively degrade signal quality and limit functional lifetime[7]. These failures are compounded by reliance on single-frequency impedance measurements (1 kHz) to characterise electrodes[8], which poorly predict in vivo behaviour and obscure how electrode geometry, motion, and tissue state affect signal transmission. As a result, the field lacks a mechanistic understanding of how bioelectrical signals are generated, transmitted, and maintained at stable interfaces. Consequently, advances in mechatronic prostheses remain underutilised, pending the development of effective, long-term closed-loop biological interfaces.

Challenges

High-quality, stable **bioelectrical interfacing is constrained by a fundamental mechanical mismatch**: biological tissues are soft, dynamic, and moving, while conventional electrodes are rigid and static. Current state-of-the-art platforms (e.g. Neuralink, Blackrock) rely on traumatic insertion strategies that cause immediate tissue damage and exacerbate micromotion-induced injury[9,10]. Motion is a critical failure mode whether in brain, nerve, or muscle, rigid implants cannot move coherently with tissue, leading to cellular shear, inflammation, and fibrotic encapsulation.

These effects result in **low electrode yield and poor long-term selectivity**, as active cells are displaced from recording sites[7]. Rather than addressing this biological interface problem, most current programs rely on increasingly complex decoding algorithms to compensate for signal loss. However, decoding intention alone is insufficient. Functional restoration requires **closed-loop control: structured stimulation to drive movement and sensory feedback to refine it**[5]. Delivering spatially and temporally precise stimulation in moving tissue, while simultaneously recording the resulting muscle response, remains a major unresolved challenge.

Neuromuscular junctions (NMJs) offer a less invasive and biologically meaningful access point for motor control, but current myoelectric interfaces lack the spatial resolution needed to selectively target motor units. Surface and bulk electrodes cannot support patterned stimulation or resolve fine muscle activation, and existing systems cannot remain intimate with the NMJ during contraction. **The core challenge is therefore to achieve selective, stable interfacing at the NMJ that moves with tissue and supports both recording and stimulation**.

Innovative solution and significance

This project proposes an **interface-centric paradigm shift** by targeting the NMJ as the primary gateway for closed-loop motor control. The central innovation is not simply signal acquisition but resolving the biological and mechanical incompatibility that limits existing interfaces. By **identifying where and how to interface with muscle to achieve selectivity**, this project enables rational electrode design rather than empirical trial-and-error.

Building on my expertise in three-dimensional tissue models and impedance monitoring[11,12], together with flexible carbon fibre electrodes[13–16], this project will establish a three-dimensional NMJ platform integrated with thread-like electrodes designed to move with tissue (**Fig. 1**). Computational simulations will first identify optimal electrode geometry, placement, and orientation for selective innervation and signal fidelity. These predictions will then directly inform electrode fabrication and experimental validation.

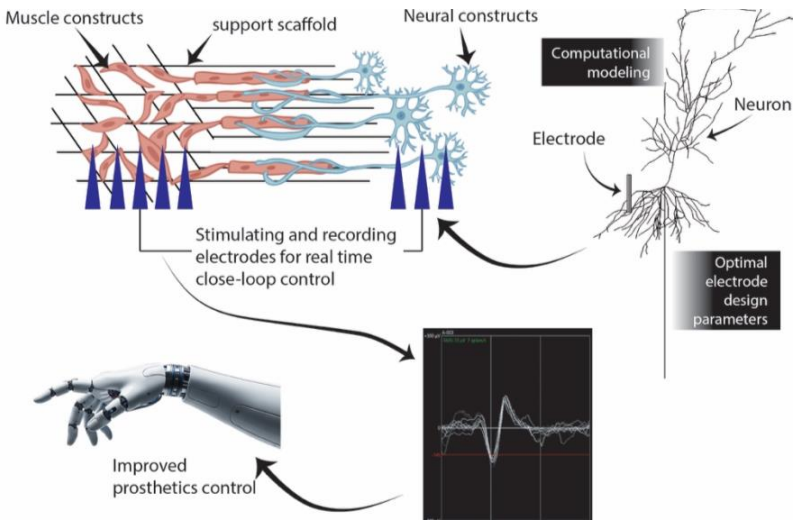

Figure 1. Computational and 3D culture modelling of electrodes to inform optimal design parameters of body-machine interfaces in prosthetics control.

By combining selective stimulation of nerves with high-resolution recording from muscle, this platform enables direct measurement of how structured stimulation patterns drive contraction. Incorporating feedback from muscle activity closes the control loop, allowing intention to be refined into

coordinated movement. **This work delivers a safe, stable, and biologically grounded interface capable of long-term operation**, enabling high-fidelity closed-loop prosthetic control and greater agency for users.

The specific **aims** of this project are:

Aim 1. To elucidate how electrode geometry, placement, and interface properties govern bioelectrical signal transduction at the neuromuscular junction.
Aim 2. To investigate scaffold architectures to establish physiologically relevant neuromuscular junction platforms for electrode evaluation.
Aim 3. To integrate optimised electrodes with 3D neuromuscular junction cultures to characterise and quantify neuron-muscle communication and evaluate closed-loop stimulation and recording strategies relevant to prosthetic control.

Research questions: The central hypothesis of this project is that key limitations in current brain-machine interfaces arise from unresolved biological and mechanical constraints at the electrode-tissue interface, rather than from deficiencies in decoding algorithms. Specifically, this project theorises that studying bioelectrical signal generation, transmission, and feedback at the NMJ will reveal how selective, stable, and motion-tolerant interfaces can be achieved - enabling closed-loop control that more closely reflects natural movement intention. The project therefore addresses the following research questions:
(i) How do electrode geometry, placement, and mechanical compliance govern selective signal capture and stimulation at the NMJ, particularly in moving tissue?

(ii) Where within muscle tissue should electrodes interface to achieve selective innervation of motor units while minimising crosstalk and signal degradation?

(iii) How do tissue structure, maturation state, and electrode-tissue coupling influence extracellular signal fidelity, stability, and responsiveness to structured stimulation?

(iv) Can muscle-based recordings be used as a reliable feedback signal to quantify and refine stimulation-driven movement in a closed-loop system?

To address these questions, the project will establish a digital twin framework that combines NEURON-COMSOL simulations with physiologically relevant three-dimensional NMJ cultures. Simulations will be used to predict optimal electrode geometry and placement for selective interfacing, which will then be experimentally validated. This integrated framework will provide a mechanistic understanding of how to design compliant electrodes that move with tissue, support patterned stimulation, and enable feedback-driven control-providing a transferable platform for next-generation body-machine and prosthetic interfaces.

Project design and methodology

Aim 1. To elucidate how electrode geometry, placement and interface properties influence bioelectrical signal transduction at the neuromuscular junction

Preliminary data: I have developed carbon fibre electrodes (**Fig. 2a**) that are now used in brain-machine interface applications[14–16] and have modelled optimal electrode geometries for high-quality neuronal recording (**Fig. 2c**). This work clarified key design principles governing signal acquisition at the electrode-tissue interface. While these electrodes can capture high-quality single-unit recordings (**Fig. 2b**), achieving high-resolution, chronic neural recording with penetrating electrodes remains challenging, limiting decoding fidelity, long-term stability, and overall device performance. Importantly, even with ideal electrode functionality, current interfaces substantially undersample neural networks, underscoring the need to maximise the information extracted from each individual electrode rather than relying solely on increased electrode count.

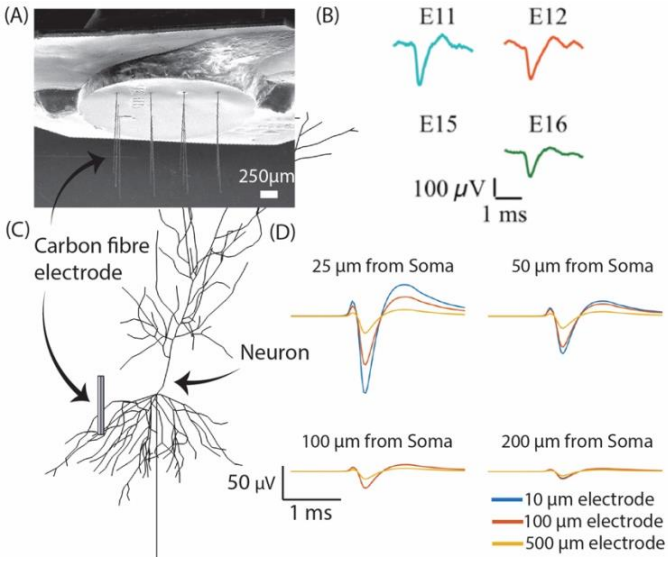

Figure 2. (A) Carbon fibre array can record (B) in vivo single unit action potentials in the visual cortex of rats and (C) COMSOL simulations allow studying the effect of (D) microwire electrodes size and distances from neuron.

My work focuses on optimising the interaction between individual electrodes and their local biological environment. By refining electrode geometry, exposed surface area, and placement relative to specific neural or muscle targets, signal fidelity at each recording site can be maximised. Computational modelling, combined with

targeted experimental validation, provides a powerful framework to predict and test how these design parameters influence recording performance. This iterative simulation-experiment loop enables systematic validation of electrode designs and increases confidence in translating predictive models into reliable, long-term bioelectronic interfaces.

Recently, together with my Capstone students, I developed a COMSOL-based modelling framework that identifies optimal conditions for microwire-like carbon fibre electrodes to record extracellular signals from nearby neurons (**Fig. 2d**). This framework has been released as open-source code on GitHub (https://github.com/sodeve19/NEURON_in_COMSOL), and a manuscript describing the results is currently in preparation, with myself as both last and corresponding author. The simulations highlight the critical importance of electrode proximity, demonstrating that positional differences on the order of 50 µm can produce substantial changes in recorded signal amplitude. Notably, the framework enables high-fidelity, coupled simulations of neuronal activity and electrode behaviour, providing a generalisable tool to inform electrode design across a broad range of neural interface applications.

Research method:

1.1. Development of model in COMSOL

Building on my established expertise in modelling the neuron-electrode interface, I will extend this framework to myotubes - differentiated muscle fibres formed from myoblasts. While the NEURON simulation framework includes open-source electrophysiological models for myoblast activity, these models can be translated into COMSOL to enable coupled electrical and physical simulations. This adaptation will allow systematic investigation of how electrode positioning relative to the myotube axis influences extracellular signal amplitude, as well as how electrode geometry affects signal-to-noise ratio and spatial selectivity.

I will investigate a range of electrode configurations, spanning surface electrodes and penetrating electrodes. Surface electrodes will be modelled in proximity to the tissue, whereas penetrating electrodes will be placed within the tissue, oriented either parallel or perpendicular to the myotube axis. These simulations will then be integrated with previously developed neuronal models to create a digital twin of the NMJ interface.

COMSOL's electrochemistry module will enable impedance modelling of the electrode-tissue interface. Although this module is rarely applied to biological tissue-electrode systems, it will be used here to extract mechanistic insight from the full impedance spectrum rather than relying solely on impedance magnitude at 1 kHz. By resolving the capacitive, resistive, and frequency-dependent components at the interface, this approach will identify how electrode geometry and tissue coupling shape signal transmission and energy loss across frequencies. Importantly, this analysis will inform the selection of stimulation and recording frequency ranges that maximise charge transfer efficiency while minimising tissue damage, providing a rational basis for designing stimulation and recording protocols relevant to NMJ-based prosthetic control.

1.2 Development of arrays based on optimal simulation results

The simulation results will directly inform the design and fabrication of carbon fibre electrode arrays for integration with the tissue cultures. Fabrication will build on established protocols using carbon fibre electrodes integrated with flexible polyimide printed circuit boards (PCBs). The arrays will be coated with Parylene-C, and electrode deinsulation (active site size) will be defined based on the outcomes of Aim 1.1.

The overall array architecture - including whether electrodes are oriented horizontally or vertically - will likewise be determined by the results of the preceding aims. In particular, electrode shape, size, and exposed surface area will be tailored to specific neural or muscle targets so that stimulation thresholds and voltage requirements for defined electrophysiological responses can be tuned with greater precision. This ensures that fabrication is guided by quantitatively optimised interface parameters that directly link electrode geometry to functional performance.

Following fabrication, electrodes will be characterised on the bench using controlled voltage stimulation in physiologically relevant media to validate and correlate simulation predictions with experimental measurements. The outcomes of this aim are expected to generate high-quality, publishable insights into electrode design optimisation, with relevance to brain-machine and neuromuscular interface technologies.

Expected outcomes: These simulations will extend previously developed neuronal models to define optimal electrode geometries for maximising recording signal quality from both myotubes, relevant to neuromuscular junction studies, and myoblasts, representing early differentiation states. In parallel, the modelling will identify stimulation parameters at the neuronal interface that support stable and informative closed-loop feedback. The resulting design rules will be directly transferable to the development of NMJ-based prosthetic interfaces and will enable more efficient, targeted fabrication of electrodes optimised for selective and stable NMJ interfacing.

Aim 2. To investigate scaffold architectures to establish physiologically relevant neuromuscular junction platforms for electrode evaluation

Preliminary data: My mentor, Prof. Kapsa, and his group at St Vincent's Hospital and RMIT University have extensive expertise in 3D skeletal muscle cultures[17]. We routinely culture these muscle constructs and have recently collected preliminary electrophysiological data using carbon fibre electrodes (**Fig. 3**). These initial recordings confirm that carbon fibre electrodes can reliably capture extracellular signals from 3D muscle cultures. The group has also demonstrated that melt electro-writing (MEW) scaffolds promote alignment of human skeletal muscle cells[17].

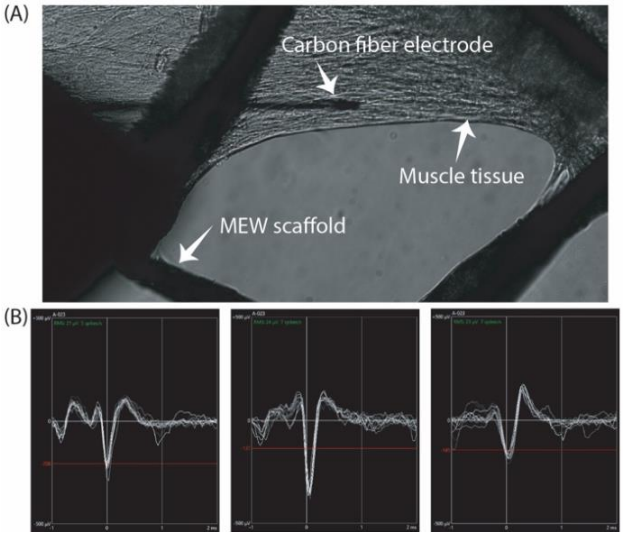

Figure 3. Carbon fibre electrodes (A) can record electrical activity from muscle tissues (B).

Furthermore, using our carbon fibre electrodes, we have shown that myotube alignment in vitro leads to enhanced muscle contractility compared to flat substrates, likely due to improved cellular alignment, maturation, and functional integration within a structured 3D environment. I have worked closely with the team to establish and optimise the electrophysiology platform used for these recordings (**Fig. 3b**), demonstrating the ability to reliably capture multiple single-unit waveforms from the muscle constructs. This platform can be further extended to support electrical stimulation and to investigate signal propagation along aligned myotube cultures.

Research method:

2.1. Evaluation of stimulation and recording parameters in aligned muscle cultures

Using established 3D muscle cultures developed by my collaborators, together with the optimised electrode designs derived from Aim 1, this work will systematically investigate how electrode geometry, placement, and stimulation parameters influence signal propagation and recording fidelity in aligned myotube constructs. Once myotubes are mature and aligned, carbon fibre electrodes will be used to deliver controlled electrical stimulation while simultaneously recording extracellular responses along the muscle fibres.

By analysing signal propagation velocity, spatial selectivity, and signal-to-noise ratio, this platform will enable quantitative assessment of optimal stimulation amplitudes, pulse widths, and recording configurations. These experimental results will be used to directly validate and refine the computational electrode models developed in Aim 1, closing the simulation-experiment loop. Importantly, cellular responses to stimulation will also be monitored to assess potential effects on muscle viability and function, informing parameter ranges suitable for stable, long-term interfacing and downstream signal decoding.

2.2. Integration of optimised electrodes for neuromuscular junction formation

Building on the validated stimulation and recording parameters, MEW scaffolds will be used to integrate aligned muscle constructs with motor neurons to form functional neuromuscular junctions. Scaffold architectures will be designed to guide axonal growth toward targeted muscle regions, enabling precise electrode placement relative to synaptic sites.

Initial studies will use animal-derived motor neurons and myoblasts to allow rapid optimisation and validation. This co-culture system will enable direct investigation of how optimised electrode parameters support bidirectional communication-stimulating neurons and recording muscle responses-under controlled, physiologically relevant conditions. The resulting NMJ platform provides a critical testbed for evaluating electrode performance in closed-loop neuromuscular interfacing scenarios relevant to prosthetic control.

Expected outcomes: In the future, the models developed here have strong potential for informing the design and optimisation of mechatronic and bionic devices that rely on stable neuromuscular interfacing. By capturing NMJ behaviour in controlled 3D systems, the platform can also be applied to conditions where NMJ function is altered, including muscular dystrophies and neurogenic muscle disorders. The integration of quantitative electrophysiology, impedance monitoring, and controlled 3D NMJ geometry will enable high-resolution analysis of signal transmission, interface stability, and functional degradation, supporting both device development and evaluation of intervention strategies.

Aim 3. To integrate optimised electrodes with 3D neuromuscular junction cultures to characterise and quantify neuron-muscle communication and evaluate closed-loop stimulation and recording strategies relevant to prosthetic control

Research method: This aim will integrate the optimised electrode parameters established in Aim 1 with the physiologically relevant muscle and NMJ platforms developed in Aim 2. A compartmentalised device incorporating custom MEW scaffolds will be developed to support aligned 3D skeletal muscle constructs while enabling controlled interaction with motor neurons cultured in an adjacent neuronal compartment (**Fig. 4**). This architecture enables precise spatial control of electrode placement relative to axons, synapses, and muscle fibres.

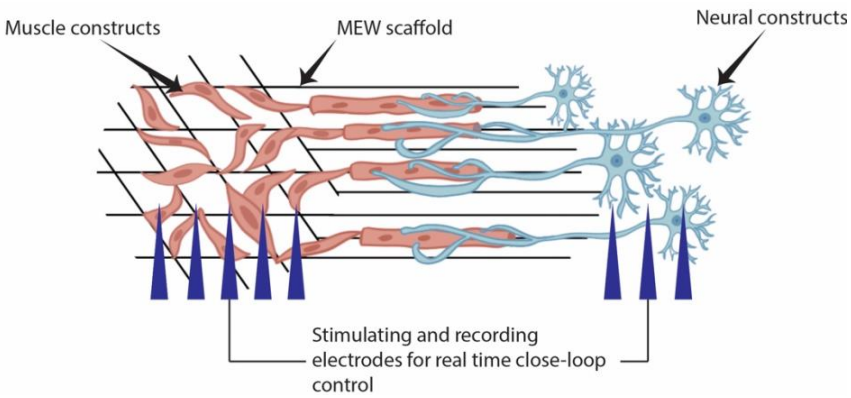

Figure 4. Neuro-muscular 3D culture construct with incorporated electrodes for recording and stimulation.

Carbon fibre electrodes designed and validated through COMSOL simulations (Aim 1) will be embedded directly within the device to enable simultaneous stimulation and recording at defined locations along the neuron-muscle pathway. Electrophysiological readouts - including extracellular action potentials, synaptic transmission fidelity, propagation dynamics, and impedance spectroscopy - will be used to quantify the stability, selectivity, and efficiency of neuron-muscle communication.

This platform will be used to investigate closed-loop stimulation and recording at the neuromuscular junction, linking defined stimulation patterns to measurable muscle responses. By systematically varying stimulation and recording configurations, the device will enable evaluation of electrode designs capable of supporting bidirectional communication at the NMJ, providing experimentally grounded insight into interface strategies suitable for future prosthetic control systems.

Expected outcomes: This aim will deliver the first integrated experimental platform that combines optimised, tissue-compliant carbon fibre electrodes with physiologically relevant 3D neuromuscular junction cultures. The work will generate quantitative benchmarks describing how stimulation patterns applied at defined neuronal locations translate into measurable muscle responses under controlled conditions.

Specifically, the outcomes will include: (i) validated relationships between electrode geometry, placement, and stimulation parameters and the resulting neuron-muscle signal fidelity and stability; (ii) experimental evidence of how closed-loop stimulation and recording strategies can be implemented at the NMJ using flexible, high-resolution interfaces; and (iii) quantitative measures of signal propagation, synaptic transmission efficiency, and impedance behaviour during sustained interfacing.

Together, these outcomes will establish a reproducible framework for assessing electrode performance in a dynamic neuromuscular environment and provide experimentally grounded design rules for interfaces intended to support bidirectional communication at the NMJ. The resulting datasets and device architecture will form a reference platform for future studies of neuromuscular interfacing and next-generation body-machine interface development.

Project feasibility: This project is well positioned to commence immediately, supported by strong preliminary data demonstrating both the feasibility of simulating biological cells and the ability of carbon fibre electrodes to record extracellular signals from muscle tissue. This preliminary work was generated through my own research in collaboration with Prof. Kapsa's group, as well as through the successful supervision of Capstone projects.
The project takes advantage on my expertise in 3D cell culture, biomaterials, and electrode fabrication and modelling-skills that are fundamental to successful delivery of the proposed aims. My capacity to complete and oversee the project is further supported by a strong track record in supervising students through to successful completion. The project is structured around three integrated aims. One PhD student will work closely with me on the development of computational simulations and electrode fabrication for Aim 1, while Aim 2 will be supported by established collaborators and another PhD student. This PhD students will receive comprehensive training spanning computational modelling, device fabrication, and bioelectronic interfacing and 3D cell culture, positioning them for a strong career in the rapidly growing field of biomedical devices and brain-machine interfaces.

Table 1. Timeline for this DECRA project. (I: DECRA applicant, P: PhD student, C: Collaborators)

Aims	I	P	C	2027				2028				2029			
Aim 1. Elucidate how electrode geometry, placement and interface properties influence bioelectrical signal transduction at the NMJ															
1.1. Development of model in COMSOL	✓	✓													

1.2. Development of Array based on optimal simulation results	✓	✓					■	■	■	■	■				
Aim 2. Investigate scaffold architectures to establish physiologically relevant NMJ platforms for electrode evaluation															
2.1. Evaluation of stimulation and recording parameters in aligned muscle cultures	✓		✓	■	■	■	■								
2.2. Integration of optimised electrodes for neuromuscular junction formation	✓		✓			■	■	■	■	■	■				
Aim 3. To integrate optimised electrodes with 3D neuromuscular junction cultures to characterise and quantify neuron-muscle communication and evaluate closed-loop stimulation and recording strategies relevant to prosthetic control															
Integrating electrodes with co-cultures and gather long-term electrophysiological and impedance spectrum data	✓	✓										■	■	■	■
Dissemination	✓	✓	✓					■				■			■

Potential of the project to enhance national and international collaboration

At the national level, this project builds directly on an established collaboration with Prof. Kapsa's group at St Vincent's Hospital and RMIT University, which has internationally recognised expertise in 3D skeletal muscle cultures, MEW scaffold fabrication, and neuromuscular disease modelling. This partnership enables seamless integration of advanced biomaterial scaffolds and physiologically relevant NMJ models with the electrode fabrication and interface-modelling capabilities developed in this project. The work also builds on strong collaborations with my mentors A/Prof David Garrett and Dr Arman Ahnood in device and electrode fabrication. Shared infrastructure, co-supervision of students, and exchange of experimental platforms will strengthen Australia's translational research capacity at the interface of bioengineering and neuroscience, particularly as this project will be based at the Aikenhead Centre for Medical Discovery (ACMD). In addition, my ongoing collaboration with CarbonCybernetics, Prof. Steven Prawer and Dr. Wei Tong at the University of Melbourne provides complementary expertise in device and materials fabrication, further enhancing the project's national collaborative framework.

At the international level, the project leverages open-source computational frameworks and modelling approaches that are widely used by the global neuroengineering community. The COMSOL-NEURON modelling pipeline developed in this work builds on, and contributes back to, existing international efforts in biophysical modelling of neural interfaces. By releasing simulation code, electrode designs, and impedance-analysis workflows through open repositories, this project will facilitate reproducibility and uptake by international groups working on neural, peripheral nerve, and neuromuscular interfaces. This open-science strategy lowers barriers to collaboration and positions Australia as a contributor of enabling technologies rather than a passive adopter.

The interdisciplinary nature of this project: combining computational simulations, electrode design, impedance spectroscopy, and three-dimensional neuromuscular junction models, creates clear opportunities for future collaboration with international groups working in prosthetics, rehabilitation engineering, neuromuscular biology, and implantable bioelectronics. The platform developed here is readily extensible to patient-derived induced pluripotent stem cell motor neurons and myotubes, providing a natural entry point for clinically oriented collaborations in neurology and regenerative medicine. In the longer term, the design rules and experimental platforms generated by this work will support partnerships with both academic and industry stakeholders seeking alternatives to invasive brain implants for prosthetic control. By delivering a validated, modular, and biologically grounded NMJ interface platform, the project establishes a strong foundation for sustained national and international collaboration in next-generation human–machine interfaces and positions Australia as a leader in interface-centric prosthetic research.

BENEFIT

New knowledge: This research will generate new mechanistic knowledge on how bioelectrical signals are generated, transmitted, and recorded at the tissue–electrode interface, with a particular focus on neuromuscular junctions (NMJs). By moving beyond conventional single-frequency impedance reporting, the project will establish how the full impedance spectrum relates to electrode geometry, material properties, and tissue state in both neural and muscular systems. This will advance fundamental understanding of the electrochemical and biophysical processes that govern signal fidelity, stability, and longevity at implantable interfaces. In addition, the coupled NEURON-COMSOL simulations will establish a new framework for creating digital twins of electrode-culture interfaces.
The project will deliver the first systematic, quantitatively validated design rules for optimising electrode geometry and placement for recording from myotubes and myoblasts, bridging a critical knowledge gap between neuronal interface

theory and neuromuscular physiology. These insights will clarify how electrode orientation, spatial selectivity, and mechanical coupling influence extracellular signal amplitude and signal-to-noise ratio during muscle differentiation and synaptic transmission at the NMJ.

In addition, the development of physiologically relevant 3D NMJ models integrated with carbon fibre electrodes will provide new knowledge on functional neuron-muscle communication under controlled conditions to inform close-loop control of prosthetics. This platform will enable direct measurement of synaptic transmission fidelity, chronic recording stability, and interface degradation-parameters that are difficult or impossible to isolate in vivo. The resulting datasets will establish reference benchmarks for NMJ electrophysiology and impedance behaviour that are currently absent from the literature.

Benefit for Australian and international communities: This research aligns directly with Australia's National Manufacturing Priority in Medical Products and the MedTech and Pharma Growth Centre Roadmap, which emphasise advanced medical devices, implantable technologies, and translation of high-value research into clinical and commercial outcomes. By addressing fundamental limitations in electrode yield, stability, and interface reliability, the project provides enabling knowledge critical to the development of next-generation prosthetic and bioelectronic devices.

The design rules and interface-optimisation framework generated by this work will support Australian medical device manufacturers, neurotechnology startups, and rehabilitation engineering companies by reducing development risk, fabrication cost, and failure rates associated with implantable interfaces. Improved electrode efficiency and longevity have the potential to lower downstream healthcare costs by reducing revision surgeries and device replacement, a key consideration for sustainable healthcare delivery. From a social and health perspective, the project supports the development of less invasive prosthetic control technologies, improving safety, patient acceptance, and long-term usability for Australians living with limb loss or neuromuscular impairment. The 3D NMJ models developed in this project will also provide new platforms for studying neuromuscular diseases such as muscular dystrophy and motor neuron disease, supporting preclinical testing and reducing reliance on animal models.

Internationally, the project positions Australia as a contributor on the growing and exciting field of brain-machine interfaces and enabling interface technologies rather than end-stage devices, strengthening global partnerships in neuroengineering and bioelectronics. This capability-building approach directly supports Australia's strategy to grow a globally competitive, innovation-driven MedTech manufacturing sector.

COMMUNICATION OF RESULTS

Digital models generated through this project will be available in open-source platforms and published in high-ranking internationally recognised journals in the field and in accordance with the ARC's Open Access policy. Due to the multidisciplinary scope of the project, the results will be suited for journals such as *Advanced Functional Materials, Nature Biomedical Engineering, Nature Communications and Advanced Healthcare Materials.* Data will be presented in national and international conferences including EMBC, FENS, and ABEC. Results will be also shared within seminars at RMIT and ACMD, and will be disseminated for the public using RMIT website and LinkedIn.

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1. 10 Brain-Computer Interface Companies Now Testing in Humans. https://www.neurofounders.co/articles/ten-bci-companies-now-in-human-trials (2025). **2.** Drew, L. A brain implant that could rival Neuralink's enters clinical trials. *Nature* 648, 14–15 (2025). **3.** Elon Musk's Neuralink implants brain chip in first human | Reuters. https://www.reuters.com/technology/neuralink-implants-brain-chip-first-human-musk-says-2024-01-29/ (2024). **4.** Greenspon, C. M. *et al.* Evoking stable and precise tactile sensations via multi-electrode intracortical microstimulation of the somatosensory cortex. *Nature Biomedical Engineering 2024 9:6* 9, 935–951 (2024). **5.** Hughes, C., Chen, X., Grill, W. & Kozai, T. D. Y. Neural mechanisms underlying intracortical microstimulation for sensory restoration. *Nature Biomedical Engineering 2026 10:2* 10, 197–213 (2026). **6.** Topalovic, U. *et al.* A wearable platform for closed-loop stimulation and recording of single-neuron and local field potential activity in freely moving humans. *Nat. Neurosci.* 26, 517–527 (2023). **7.** Preszler, C., Stieger, K., Chen, K., Zhang, G. & Kozai, T. D. Y. Microglia surveillance is directed toward neuron activation during sustained intracortical microstimulation. *J. Neural Eng.* https://doi.org/10.1088/1741-2552/AE4652 (2026) doi:10.1088/1741-2552/AE4652. **8.** Hazelgrove, B. *et al.* Electrochemical impedance spectroscopy in vivo for neurotechnology and bioelectronics. *Nature Reviews Electrical Engineering 2025 2:2* 2, 110–124 (2025). **9.** Thielen, B. & Meng, E. A comparison of insertion methods for surgical placement of penetrating neural interfaces. *J. Neural Eng.* 18, 041003 (2021). **10.** Neuralink implant limitations | The race to build state of the art implantable brain tech. https://www.paradromics.com/insights/neuralink-implant. **11.** **De Leon, S. E.**, Cleuren, L., Oo, Z. Y., Stoddart, P. R. & McArthur, S. L. Extending In-Plane Impedance Measurements from 2D to 3D Cultures: Design Considerations. Bioengineering 2021, Vol. 8, Page 11 8, 11 (2021). **12.** **De León, S. E.**, Pupovac, A. & McArthur, S. L. Three-Dimensional (3D) cell culture monitoring: Opportunities and challenges for impedance spectroscopy. *Biotechnology and Bioengineering* vol. 117 1230–1240 Preprint at https://doi.org/10.1002/bit.27270 (2020). **13.** **De León, S. E.**, Higham, S., Jung, Y. J., Tong, W. & Garrett, D. J. Recent developments in microwire-structured intracortical electrode arrays for brain–machine interfaces. *Bioeng. Transl. Med.* https://doi.org/10.1002/BTM2.10742 (2024) doi:10.1002/BTM2.10742. **14.** Edalati, K., **De León, S. E.** *et al.* Effective removal of metal-carbide contamination from diamond circuit boards by reactive ion etching. *Diam. Relat. Mater.* 139, (2023). **15.** [PREPRINT] Higham, S., **De León, S. E.** *et al.* Carbon Cybernetics Array: a miniaturized carbon-based microelectrode array for intracortical recording. *bioRxiv* 2026.01.06.696090 (2026) doi:10.64898/2026.01.06.696090. **16.** Higham, S., **De León, S. E.** *et al.* Boron-Doped Nano-Crystalline Coated Carbon Fibers for Phasic Dopamine Sensing. *Adv. Healthc. Mater.* https://doi.org/10.1002/ADHM.202503945 (2025) doi:10.1002/ADHM.202503945. **17.** Snow, F. *et al.* High resolution melt electro-written scaffolds promote alignment of human skeletal muscle cells. *Biofabrication* 17, 035013 (2025).

C2. Administering Organisation Letter of Support

(Provide a Letter of Support from the Administering Organisation confirming its commitment to the appointment of the participant, and the application. Refer to the Instructions to Applicants for the submission requirements. (Upload a PDF of up to 2 A4 pages).)

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Part D - Project Cost (DE270101085)

D1. What is the proposed budget for your project?

(There are rules around what funds can be requested from the ARC. You must adhere to the scheme specific requirements listed in the Grant Guidelines. Refer to the Instructions to Applicants for detailed instructions on how to fill out the budget section.)

Total requested budget: $529,216

Year 1

Description	ARC	Admin Org
	Cash	Cash
Total	179,072	100,722
Personnel	164,072	70,992
Dr Sorel De Leon Vergara (Discovery Early Career Researcher Award)	129,734	31,405
HDR (Higher Degree by Research stipend)	34,338	2,624
HDR (Higher Degree by Research stipend)		36,963
Travel		4,730
Total travel costs		4,730
Maintenance	15,000	25,000
Total maintenance costs	15,000	25,000

Year 2

Description	ARC	Admin Org
	Cash	Cash
Total	179,072	115,116
Personnel	164,072	76,616
Dr Sorel De Leon Vergara (Discovery Early Career Researcher Award)	129,734	37,029
HDR (Higher Degree by Research stipend)	34,338	2,624
HDR (Higher Degree by Research stipend)		36,963
Travel		13,500
Total travel costs		13,500
Maintenance	15,000	25,000
Total maintenance costs	15,000	25,000

Year 3

Description	ARC	Admin Org
	Cash	Cash
Total	171,072	115,733
Personnel	164,072	82,233
Dr Sorel De Leon Vergara (Discovery Early Career Researcher Award)	129,734	42,646
HDR (Higher Degree by Research stipend)	34,338	2,624

HDR (Higher Degree by Research stipend)		36,963
Travel		13,500
Total travel costs		13,500
Maintenance	7,000	20,000
Total maintenance costs	7,000	20,000

D2. Justification of funding requested from the ARC

(Fully justify, in terms of need and cost, each budget item requested from the ARC. Use the same headings as in the Description column in the Budget Table of this application (Upload a PDF of up to 3 A4 pages and within the required format).)

Budget Justification

Uploaded PDF file follows on next page.

D2 JUSTIFICATION OF FUNDING REQUESTED FROM THE ARC

PERSONEL

HDR (higher Degree by Research stipend): $34,338 in Year 1, Year 2 and Year 3

This project will support the training of a full-time PhD student in the brain-machine interface field, who will play a central role in delivering the proposed research. The student will be trained by Dr De Leon in data acquisition and analysis, electrode fabrication, electrochemistry, and cell culture, and will develop advanced skills in analysing physiological recordings. The PhD student will be instrumental in executing key components of the project, particularly Aim 1 and Aim 3, which require sustained experimental and analytical effort to integrate sensing functionality into microelectrodes and demonstrate their suitability for long-term neural interfacing. While the project objectives are realistic and well defined, successful and timely completion depends on the contribution of a dedicated HDR candidate. Through this training, the student will graduate with highly specialised, industry-relevant expertise in neural interfacing technologies.

MAINTENANCE

COMSOL Multiphysics license: $5,000 in Year 1 and Year 2

A COMSOL Multiphysics licence, including the electrochemistry and electrical modules, is essential for delivery of Aim 1, as it provides the simulation platform required to model electrode–tissue interactions. COMSOL enables coupled electrochemical and physics-based simulations with predefined physicochemical properties, which is critical for accurately capturing bioelectrical interface behaviour. The software is widely used to model electrical, chemical, and transport phenomena in biological systems and is therefore indispensable for this project.

RACE support: $3,000 in Year 1 and Year 2

While RMIT provides access to the RACE high-performance computing platform, usage is charged on a time basis. The computational simulations proposed in this project require substantial processing power and will therefore be conducted on the RACE supercomputing infrastructure. Funds of $3,000 in Years 1 and 2 are requested to cover the costs associated with running these simulations on the platform.

Electrode and scaffold fabrication: $1,000 in Year 1, 2 and Year 3.

Carbon fibres, silver epoxy, and electronic components such as cables and custom PCBs are required for the fabrication of microelectrode arrays used in this project. In addition, polymers for melt electro-writing (MEW) scaffolds, including polycaprolactone (PCL) and medical-grade PEVA (40 wt.%), will be purchased from established suppliers such as Sigma-Aldrich. These materials are essential for device fabrication and scaffold production across the project. Funding of $1,000 per year is requested to cover these consumable fabrication costs.

Cell culture consumables and reagents: $4,000 in Year 1, Year 2 and Year 3.

I will conduct cell culture experiments to develop physiologically relevant neuromuscular junction co-cultures and to investigate how neuronal-muscle communication responds to electrical stimulation. These studies will provide insight into signal transmission mechanisms relevant to body-machine interface control. The experiments require standard cell culture consumables, including sterile culture dishes, tubes, and pipettes, as well as reagents such as skeletal muscle growth medium, trypsin, DMEM, horse serum, neurobasal medium, GlutaMAX, penicillin–streptomycin, fetal bovine serum, and basic fibroblast growth factor. Reagents and consumables will be sourced from established suppliers such as Sigma-Aldrich and Thermo Fisher Scientific.

Staining chemicals: $2,000 in Year 1, Year 2 and Year 3

Immunostaining will be used for intracellular staining of the live cells to obtain the cell type and morphology. Both primary and secondary antibodies will be required to reveal neurons (anti-**β** III tubulin, NeuN), myoblast and myotubes (desmin, MyHC) post fixation of the tissue samples. These antibodies will be sourced from Abcam. These dyes will be sources from Sigma or AbCam.

D3. Details of non-ARC contributions

(Provide an explanation of how non-ARC contributions will support the proposed project. Use the same headings as in the Description column in the above Budget Table of this application. (Upload a PDF of up to 2 A4 pages and within the required format).)

Details of Non-ARC Contributions

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D3 DETAILS OF NON-ARC CONTRIBUTIONS

PERSONEL

Dr. De Leon Vergara (Discovery Early Career Researcher Award): $31,405 in Year 1, $37,029 in Year 2, and $42,646 in Year 3

This will cover the salary gap between ARC DECRA salary and RMIT University salary scale B3 (including on costs) for Dr. De Leon Vergara.

HDR (higher Degree by Research stipend): $2,624 in Year 1, $ 2,624 in Year 2 and $2,624 in Year 3

This covers PhD stipend top-up to bring the stipend to the RMIT University rates. This student will focus on expanding the simulations to neuromuscular junctions and studies the effects of electrode geometry and positioning.

HDR (higher Degree by Research stipend): $36,963 in Year 1, $ 36,963 in Year 2 and $36,963 in Year 3

RMIT will cover stipend for an additional PhD stipend at the RMIT University rates. This student will focus on the 3D models of the neuromuscular junction and understanding the effects of stimulating and recording in the system.

TRAVEL

Travel to conference (ABEC): $4,730 in Year 3

RMIT University will cover the costs for the PhD student and Dr De Leon Vergara to attend the national bioengineering conference in Year 1. This will enable dissemination of early project findings and provide opportunities to engage with researchers across Australia, strengthening national collaboration and increasing the visibility of the research within the bioengineering community. This includes conference fees ($400x2), accommodation and incidentals ($330 per day for 5 days x2), and flights ($300 to AU city tbd. x2)

Travel to conference (FENS): $13,500 in Year 2

RMIT University will support attendance for the PhD student and Dr De Leon Vergara at an international conference in Year 2. Participation in the Federation of European Neuroscience conference will enable dissemination of results generated from the simulation studies, providing an opportunity to present findings and engage with leading researchers in neuroscience and brain-machine interface research. This will strengthen international collaboration and increase the visibility and impact of the project outcomes. This includes conference fees ($1000x2), per diem ($275 per day for 5 days x2), hotel fees ($275 per day for 5 days x2) and flights ($3000 to Europe x2)

Travel to conference (EMBC): $13,500 in Year 3

RMIT will cover the costs for the PhD student and Dr De Leon Vergara to attend an international conference in Year 3. This will allow us to present our findings and communicate with other researchers in the bioengineering field. This includes conference fees ($1000x2), per diem ($275 per day for 5 days x2), hotel fees ($275 per day for 5 days x2) and flights ($3000 to Europe or USA tbd. x2)

MAINTENANCE

RMMF access: $10,000 in Year 1, Year 2, and Year 3

To image and characterise our devices and cultures, we will require access to RMIT University microscope facility, as such RMIT University will contribute with access and maintenance of this space for the three years of this project.

MNRF access: $10,000 in Year 1, Year 2, and Year 3

Access to the Micro Nano Research Facility (MNRF) is essential for the fabrication of carbon fibre electrodes and arrays. All devices will be fabricated within the MNRF at RMIT University, which will provide in-kind support through access to, and maintenance of, these facilities for the full three-year duration of the project.

RACE support: $5,000 in Year 1 and Year 2

RMIT University will match ARC support for access to the RACE high-performance computing platform, ensuring that the computational simulations proposed in this project can be completed within the stipulated timeframe. Funds of $5k in Years 1 and 2 are requested to cover the costs associated with running these simulations on the platform.

Part E - Classifications and Other Statistical Information (DE270101085)

E4. To what extent does your application align with each of the following types of research activity?

(Indicate which type(s) of research activity the project best aligns with by selecting a percentage weighting for each research type. You may proportion the research across one or more categories, but the percentages must total 100%.)

Categories

Pure basic research	40%
Strategic basic research	40%
Applied research	20%

E5. Field of Research (FoR-2020)

(Select up to 3 FoR classification codes that relate to the application. Note that the percentages must total 100.)

Code	Percentage
400302 - Biomaterials	50
400399 - Biomedical engineering not elsewhere classified	25
460207 - Modelling and simulation	25

E6. Socio-Economic Objective (SEO-2020)

(Select up to 3 SEO classification codes that relate to the application. Note that the percentages must total 100.)

Code	Percentage
280110 - Expanding knowledge in engineering	60
280115 - Expanding knowledge in the information and computing sciences	40

E7. Interdisciplinary Research

(This is a 'Yes' or 'No' question. If you select 'Yes' two additional questions will be enabled:
1. Specify the ways in which the research is interdisciplinary by selecting one or more of the options below.
2. Indicate the nature of the interdisciplinary research involved. (Up to 375 characters, approximately 50 words).)

Does this application involve interdisciplinary research?

Yes

Specify the ways in which the research is interdisciplinary by selecting one or more of the options below.

Methodology
Investigatory Team

Indicate the nature of the interdisciplinary research involved. (Up to 375 characters, approximately 50 words)

The project combines electrode fabrication, materials science, computational modelling (NEURON–COMSOL), electrophysiology, and 3D cell culture. It brings together engineering and biological expertise within a collaborative team to integrate digital simulations with experimental neuromuscular models.

E8. Does the proposed research involve international collaboration?

(This is a 'Yes' or 'No' question. If you select 'Yes' you will need to answer: If the proposed research involves international collaboration, specify the countries involved.)

No

E9. If the proposed research involves international collaboration, please specify the country/ies involved.

(Commence typing in the search box and select from the drop-down list the name of the country/ies of collaborators who will be involved in the proposed project. Note that Australia is not to be listed and is not available to be selected from the drop-down list.)

E10. How many PhDs, Masters and Honours will be filled as a result of this project?

(For reporting purposes, the ARC is capturing the number of Research Students that would be involved if the application is funded. Enter the number of all student places (full-time equivalent - FTE) that will be filled as a result of this project, not just those requested in the budget for funding in the application form.)

Number of Research Student Places (FTE) - PhD

1

Number of Research Student Places (FTE) - Masters

0

Number of Research Student Places (FTE) - Honours

0

Part F - Project Eligibility (DE270101085)

F1. Medical Research

(This is a 'Yes' or 'No' question. Does this application contain content which requires a statement to demonstrate that it complies with the eligible research requirements set out in the ARC Medical Research Policy located on the ARC website?)

Yes

F2. Medical Research Statement

(Justify why this application complies with the eligible research requirements set out in the ARC Medical Research Policy located on the ARC website. Eligibility will be based solely on the information contained in this application. This is the only chance to provide justification, the ARC will not seek further clarification. (Up to 750 characters, approximately 100 words))

The aim of this proposal is to develop an integrated experimental and digital framework to advance understanding of electrode fabrication and electrode-tissue interfaces, and to use this knowledge to inform the design and interpretation of brain-machine interfaces. The tools and insights generated through this work are intended to support fundamental neuroscience and bioengineering research. Any subsequent clinical, translational, or commercial applications of the electrodes or simulation frameworks fall outside the scope of this project.

Certification

Certification by the Deputy/Pro Vice-Chancellor (Research) or their delegate or equivalent in the Administering Organisation

I certify that—

- I have read, understood, and complied with the *Discovery Program Grant Guidelines - Fellowships (2024 edition - Variation 1)*, (Grant Guidelines) and, to the best of my knowledge all details provided in this application form and in any supporting documentation are true and complete in accordance with the Grant Guidelines.
- Proper enquiries have been made and I am satisfied that the participant and the organisation listed in this application meet the requirements specified in the Grant Guidelines, including the participant having been awarded a PhD on or after 1 March 2021. Where the participant has allowable career interruptions, sufficient evidence has been provided to the Administering Organisation and based on this evidence, I certify that the participant has an award of PhD date together with an allowable period of career interruption (as listed in the Grant Guidelines) that would be commensurate with an award of PhD date on or after 1 March 2021.
- Where the participant holds a higher research degree that is not a PhD, sufficient evidence was provided to the Administering Organisation and based on this evidence, I certify that the participant's qualification meets the level 10 criteria of the *Australian Qualifications Framework Second Edition.*
- I certify that where a participant has more than one PhD, the earliest PhD has been selected and meets the eligibility requirements, including having been awarded a PhD on or after 1 March 2021.
- Upon request from the ARC, this organisation will provide evidence to support a career interruption justification in relation to the PhD award date.
- In certifying the National Interest Test statement, I have considered the requirements detailed in the Instructions to Applicants, including whether the National Interest Test statement is written in plain English and for the audience – the general public.
- The ARC reserves the right to audit any evidence on which an application is based.
- I have received confirmation that the participant will not be undertaking any Higher Degree by Research during the project activity.
- I will notify the ARC if there are changes to the participants, and the organisation listed after the submission of this application.
- The participant is responsible for the authorship and intellectual content of this application, and has appropriately cited sources and acknowledged significant contributions to it.
- To the best of my knowledge, all material personal and financial interests and Conflicts of Interest relating to parties involved in or associated with this application are disclosed to the Administering Organisation, and, if the application is successful, I agree to manage all Conflicts of Interest relating to this application in accordance with the *Australian Code for the Responsible Conduct of Research* (2018), the *ARC Conflict of Interest and Confidentiality Policy* located on the ARC website and any relevant successor documents.
- I have ensured that any additional certification requirements, applied by the ARC as a result of findings of breaches of the *Australian Code for the Responsible Conduct of Research* (2018), are provided at the grant opportunity closing date for any named participant on an application.
- I have obtained the agreement, attested to by written evidence, of all the relevant persons and organisations necessary to allow the project to proceed. This written evidence has been retained and will be provided to the ARC if requested.
- The application, including all parties involved in or associated with the application, have undergone due diligence to assess risks from foreign interference in line with the *Guidelines to Counter Foreign Interference in the Australian University Sector* (2019) located on the ARC website and developed by the University Foreign Interference Taskforce.
- This application complies with the eligible research requirements set out in the *ARC Medical Research Policy*, located on the ARC website.
- This application complies with the eligible research requirements set out in the *ARC Experimental Development Policy*, located on the ARC website.

- This application does not request funding for the same research activities, infrastructure, or Project previously funded or currently being funded through any other Commonwealth funding.
- This application complies with the requirements to manage other similar or linked research applications by the participant and management is in place to avoid duplication of Australian Government funding if all applications are funded.
- If this application is successful, I am prepared to have the project carried out as set out in this application and agree to abide by the terms and conditions of the Grant Guidelines and the relevant Commonwealth Grant Agreement.
- The project can be accommodated within the general facilities of this organisation and if applicable, within the facilities of other relevant organisations specified in this application and sufficient working and office space is available for any proposed additional staff.
- All funds for this project will only be spent for the purpose for which they are provided.
- The project cannot commence until there is an ethics plan in place.
- I consent, or where necessary, I have obtained the necessary consent(s) concerning all the parties, to this application being referred to third parties, including to overseas parties, who will remain anonymous, for assessment purposes.
- I consent, or where necessary, I have obtained the necessary consent(s) concerning all the parties, to the ARC copying, modifying and otherwise dealing with information contained in this application.
- I acknowledge, or where necessary have informed all the parties, that information from this application may be provided to other Commonwealth agencies to seek advice on national security or other matters.
- I confirm that potential risks, are factored into the proposed project and if awarded a risk management plan will be in place before the project can commence.
- To the best of my knowledge, the Privacy Notice appearing at the top of this form was drawn to the attention of the participant whose personal details are provided in the Participant section of the application.